

MKL Performance Over Time

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2025

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1 Methodology

2 dgemm

3 dgeqrf

4 dtrsm

5 DFT 2D

6 Pardiso

7 vsExp

8 vdExp



Sourcing MKL Versions

- ▶ Intel only releases from the last year on their software center.
- ▶ Releases through 2023 through Intel's apt repository
- ▶ Some 2020 releases on Archive.org
- ▶ Official MKL python wheels on pypi include dynamic libraries as far back as 2018
 - ▶ All benchmarks used dynamic linking as fewer static libraries could not be sourced



Experimental Setup

Benchmarks

- ▶ One warm-up iteration
- ▶ 16 timed iterations
- ▶ Plots show average time
- ▶ Error bars show standard deviation
- ▶ Sequential and OpenMP variants
 - ▶ 80 threads for OpenMP builds
 - ▶ 2025.0 build of Intel's OpenMP runtime used for all benchmarks

Machine

- ▶ IceLake node on TACC
- ▶ Intel Xeon Platinum 8380 (Q2'21)
- ▶ 80 cores on two sockets

Workloads sizes differ for most experiments between sequential and OpenMP builds



- ▶ All experiments (except pardiso - OpenMP)
- ▶ Performance normalized to 2018.0.3

The chart displays the relative runtime of various BLAS routines across different MKL versions. The y-axis represents the relative runtime, ranging from 0.7 to 1.4. The x-axis represents the MKL version, ranging from 2018.0.3 to 2025.3.0. The legend identifies the following routines and their corresponding line colors:

- dgemm_openmp (blue)
- dgemm_seq (green)
- vExp_openmp (red)
- vExp_seq (purple)
- vdExp_openmp (orange)
- vdExp_seq (light blue)
- dgesqr_openmp (dark blue)
- dgesqr_seq (dark green)
- dtrsm_openmp (red)
- dtrsm_seq (purple)
- fft2d_openmp (yellow)
- fft2d_seq (light blue)

The chart shows that the relative runtime for most routines is generally stable around 1.0, with some fluctuations. The vExp_seq routine shows a significant peak in runtime around 2021.2.0. The vdExp_seq routine shows a significant drop in runtime around 2020.0. The fft2d_seq routine shows a significant drop in runtime around 2020.0.

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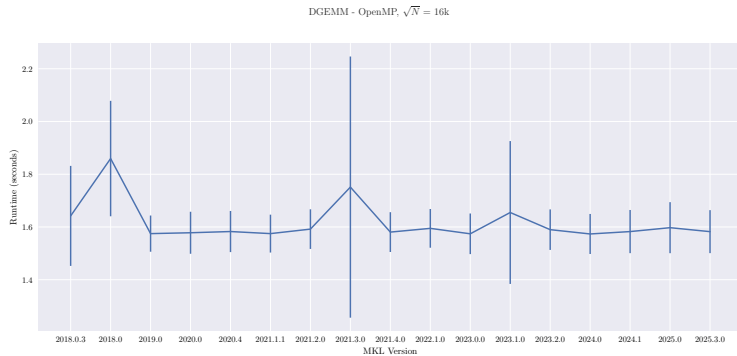
7 vsExp

8 vdExp

dgemm

- ▶ Double precision general matrix multiply
- ▶ $C = \alpha A \cdot B + \beta C$
- ▶ Less noise in sequential runtimes - expected
- ▶ Different Workload sizes,
 - ▶ $\sqrt{N} = 16000$ - OpenMP
 - ▶ $\sqrt{N} = 3200$ - Sequential

dgemm - OpenMP



dgemm - Sequential

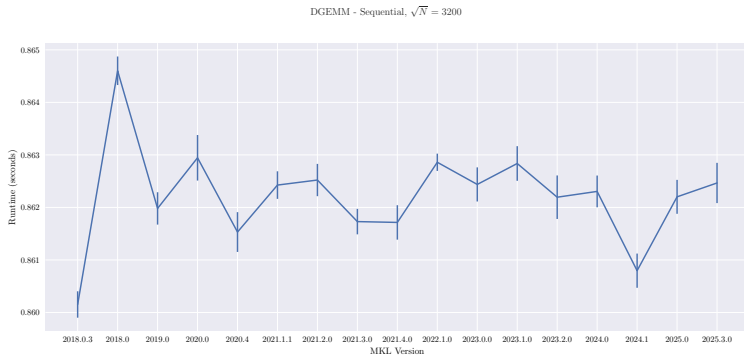


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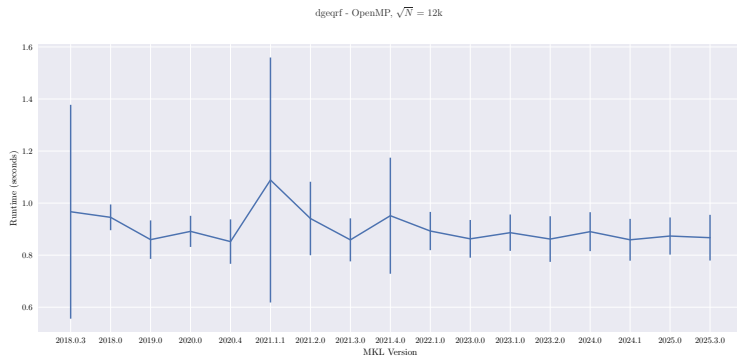
7 vsExp

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dgeqrf

- ▶ Double precision QR factorization
- ▶ Different Workload sizes,
 - ▶ $\sqrt{N} = 12000$ - OpenMP
 - ▶ $\sqrt{N} = 4000$ - Sequential

dgeqrf - OpenMP



dgeqrf - Sequential

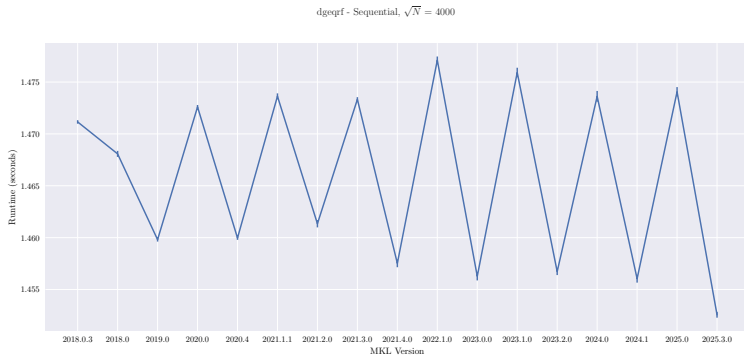


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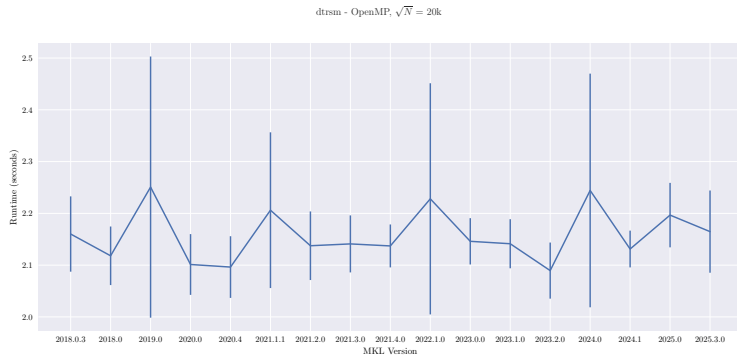
dtrsm

- ▶ Double precision solve for

$$\text{op}(A) \cdot X = \alpha \cdot B$$

- ▶ Different Workload sizes,
 - ▶ $\sqrt{N} = 20000$ - OpenMP
 - ▶ $\sqrt{N} = 5000$ - Sequential

dtrsm - OpenMP



dtrsm - Sequential

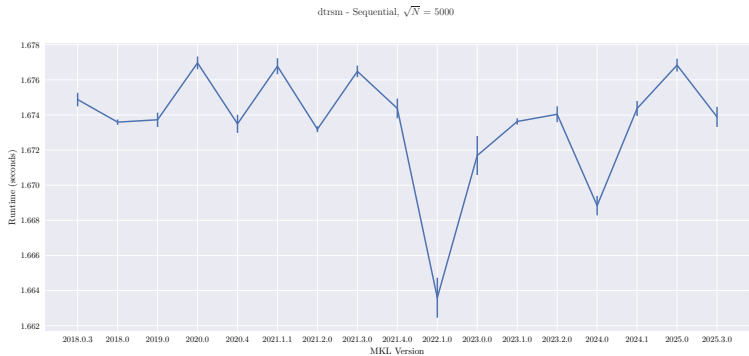


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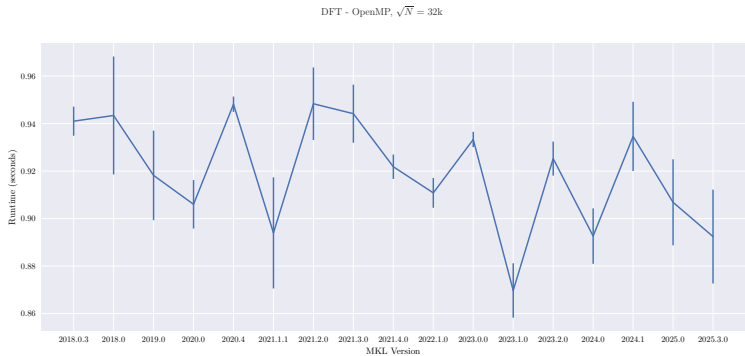
7 vsExp

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DFT 2d

- ▶ Double precision discrete Fourier transform
 - ▶ General improvement over time
- ▶ Trend is very clear for sequential
- ▶ From late 2018 release notes (build not tested)
 - ▶ Improved performance for batched real-to-complex 3D for Intel® Xeon® Processor supporting Intel® Advanced Vector Extensions 512 (Intel® AVX-512) (codename Skylake Server)
 - ▶ Improved performance with and without scaling factor across all domains.
- ▶ Different Workload sizes,
 - ▶ $\sqrt{N} = 32000$ - OpenMP
 - ▶ $\sqrt{N} = 8000$ - Sequential

DFT 2D - OpenMP



DFT 2D - Sequential

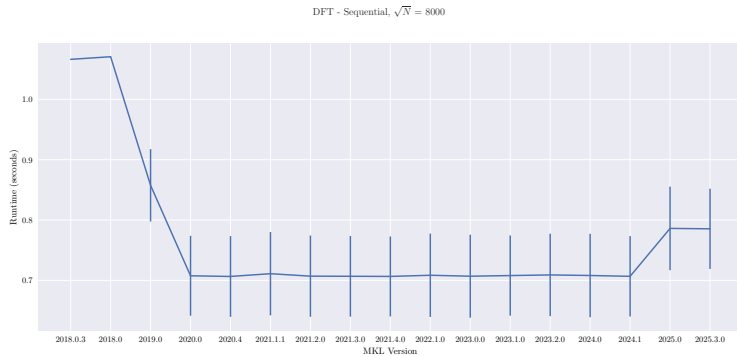


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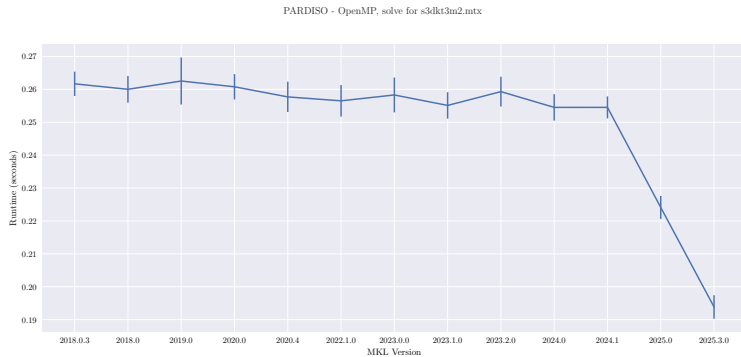
7 vsExp

8 vdExp

DFT 2d

- ▶ Double precision direct sparse linear solver
- ▶ Solve `s3dkt3m2.mtx`
- ▶ Improvement starting in 2025
- ▶ From release notes:
 - ▶ “Improved iterative refinement feature of oneMKL PARDISO”
- ▶ Same workload for OpenMP and Sequential
 - ▶ OpenMP builds for 2021 were unable to solve benchmark

Pardiso - OpenMP



Pardiso - Sequential

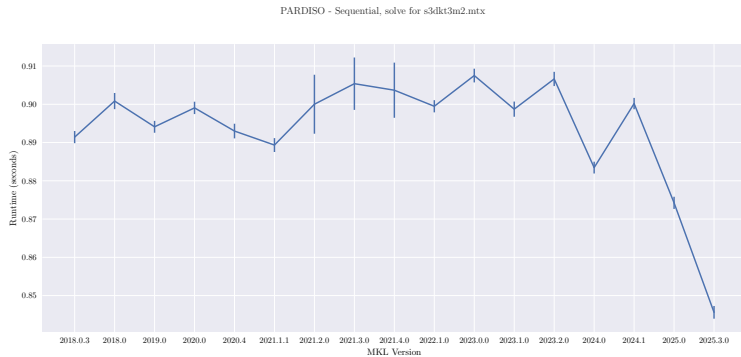


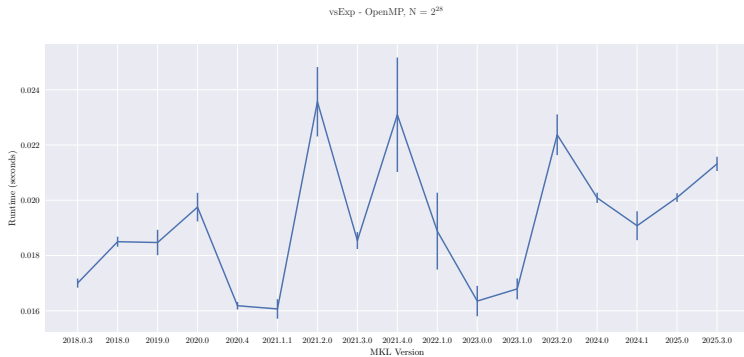
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vsExp

- ▶ Single precision element-wise exponential
- ▶ $a_i = e^{b_i}$
- ▶ Performance has degraded over time
- ▶ Same workload size for sequential and OpenMP

vsExp - OpenMP



vsExp - Sequential

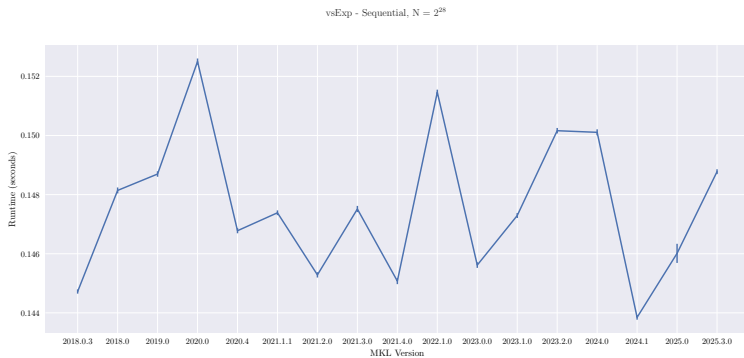


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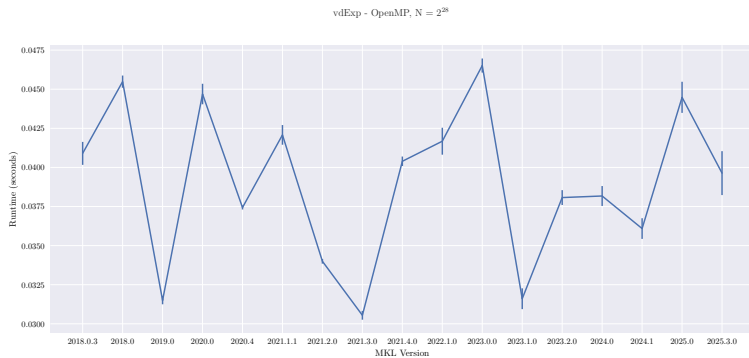
7 vsExp

8 **vdExp**

vdExp

- ▶ Double precision element-wise exponential
- ▶ $a_i = e^{b_i}$
- ▶ Same workload size for sequential and OpenMP

vdExp - OpenMP



vdExp - Sequential

vdExp - Sequential, $N = 2^{28}$ 